INTERACTIVE REVISION DECK

COMPUTER SYSTEMS

Class 10 Complete Curriculum Framework: Architecture, Hardware & Key Operational Shortcuts

TOPIC 01	TOPIC 02	TOPIC 03
IPO Framework & Logic Units	System & Memory Hubs	A-Z Operational Shortcuts

HOW COMPUTERS OPERATE

The IPO Cycle Framework

At its fundamental level, every digital computer operates under the **Input-Processing-Output (IPO)** protocol. It takes unstructured data, converts it internally, and provides actionable findings.

KEY MILESTONES

Charles Babbage originated the concept of a programmable digital device. Today, his base logical framework remains unchanged inside modern computers.

INPUT PHASE

Users translate physical instructions into digital code signals

PROCESSING PHASE

CPU logic architecture modifies, executes, and performs calculations

OUTPUT PHASE

Processed results are made readable via visual or hardcopy assets

CENTRAL PROCESSING UNIT

The System's Logical Core

The CPU serves as the central brain of a computer system. Internally, it manages instructions and coordinates data flow via three primary logical components.

ALU (Arithmetic Logic Unit)

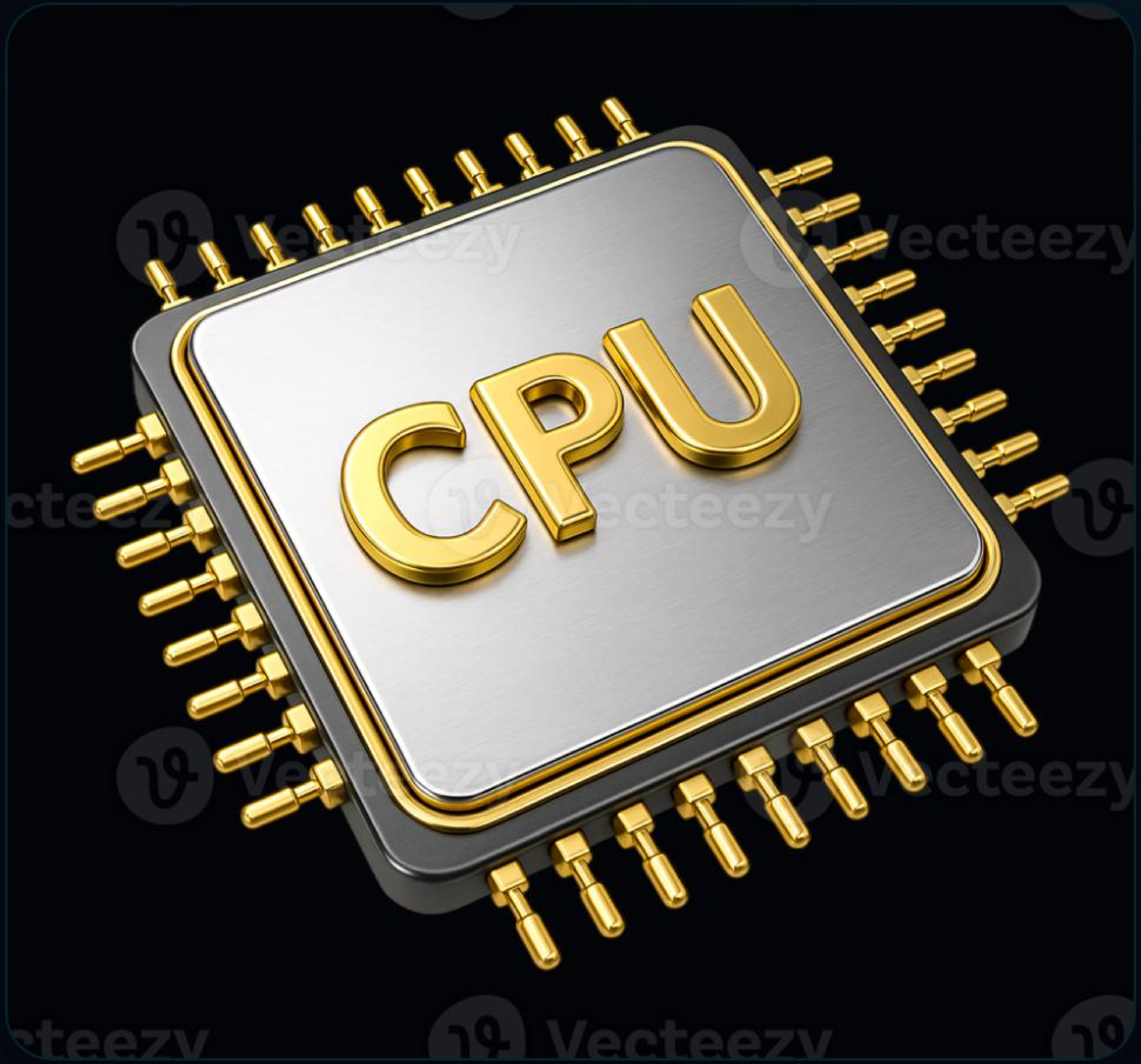
Performs all core math (addition, subtraction, division) and evaluates comparisons like <, >, or =.

CU (Control Unit)

The supervisor of internal operational pipelines. Manages execution steps and directs peripherals.

MU (Memory Unit)

Holds critical working memory datasets temporarily to ensure instant access and high calculation speed.



computer motherboard close up circuitry board

THE MAINBOARD HUB

CLASS 10 • CS

Connecting System Peripherals

Commonly referred to as the **Motherboard (MOBO)**, this large printed circuit board acts as the structural foundation of computer hardware.
All vital circuits and components connect directly here. It houses dedicated sockets for high-speed processors, memory boards, chips, expansion cards, and system ports.

RAM SLOTS
High speed physical connections to CPU

I/O PORTS
External cable plugins for input/output devices

DATA INPUT PERIPHERALS

Standard vs. Advanced

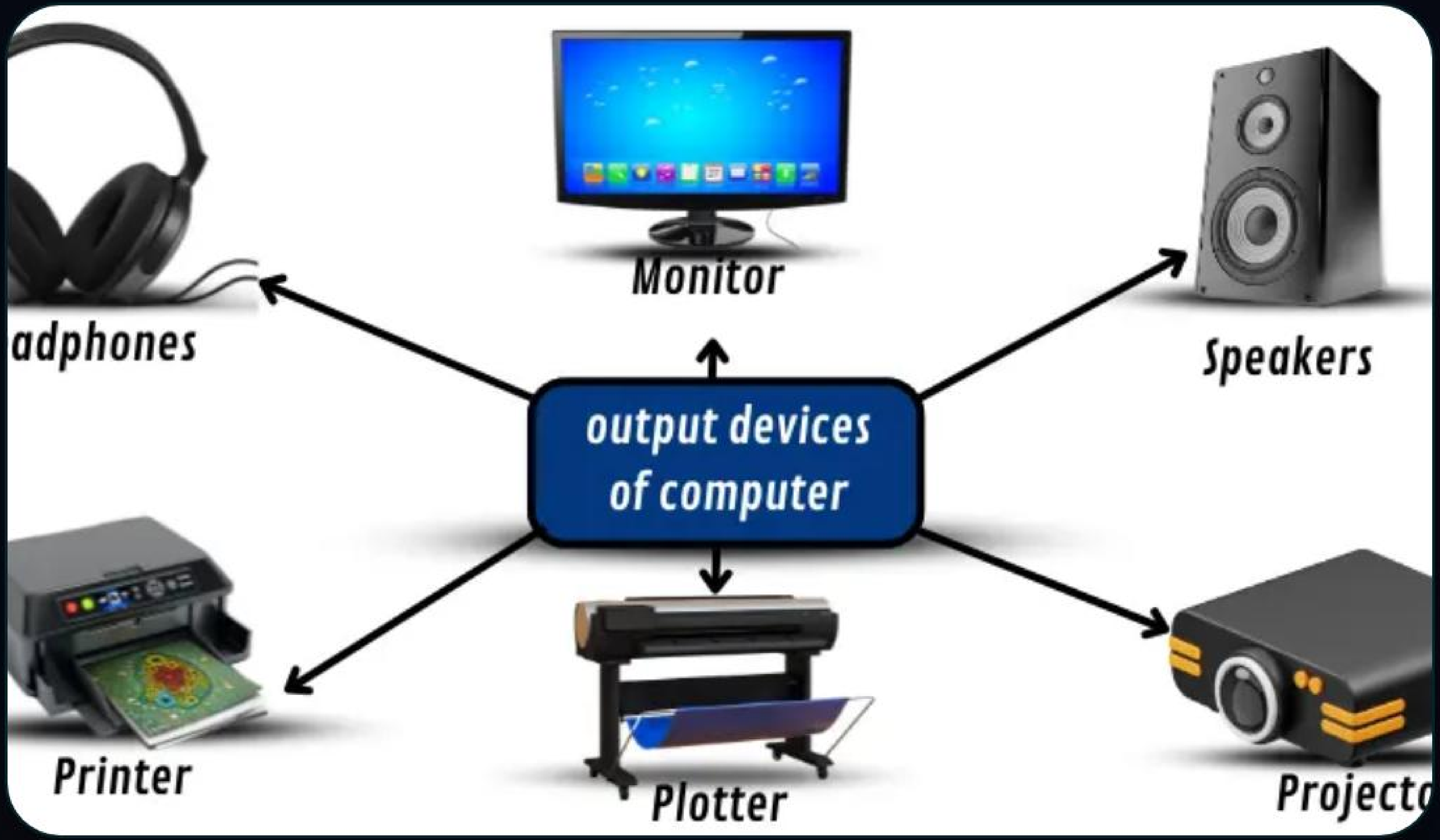
Input devices capture physical human directives or real-world parameters and convert them directly into readable binary code signals for CPU analysis.

- **Standard Instruments:** Primary data input channels like QWERTY keyboards, optical mice, and digitizers.
- **Specialized Media Devices:** Precision tools such as gaming joysticks, biometric sensors, and document flatbed scanners.
- **Interactive Screens:** Touch capacitive layers that seamlessly integrate input capabilities directly on visual display layouts.

computer input devices keyboard mouse joystick scanner

Above: Input peripherals interface users directly with operational systems.

OUTPUT SYSTEMS & MEDIA



Above: Advanced output systems deliver high quality soft and hard copies.

Soft vs. Hard Copy Output

Output hardware renders internally calculated binary files back into standard sensory formats (visual, physical paper, or sound).

- **Visual Screen Displays (Soft Copy):** Direct liquid crystal displays (LCD) or light emitting displays (LED) providing temporary information.
- **Impact & Non-Impact Printers:** Generates permanent physical documents via thermal inkjets, toner lasers, or blueprint plotters.
- **Acoustic Output Hardware:** Acoustic audio speakers and specialized headset units that play digital audio feeds.

PRIMARY MEMORY SYSTEM

Working Memory vs. Startup BIOS

Primary memory sits closest to the processor. It holds immediate software operational data to support system stability and program speed.

PERFORMANCE METRIC	RAM (RANDOM ACCESS MEMORY)	ROM (READ ONLY MEMORY)
Volatility Profile	**Volatile** — Erased when system power is disconnected.	**Non-Volatile** — Retains essential boot instructions permanently.
Access Capability	Supports rapid **Read and Write** actions on-the-fly.	Designed as **Read-Only** memory to prevent accidental loss.
Functional Purpose	Temporarily stores open system programs and operations.	Stores the foundational firmware code (BIOS / Startup program).
Speed Thresholds	Extremely fast data flow directly to core cache units.	Slightly slower compared to modern RAM registers.

SECONDARY STORAGE UNITS

Non-Volatile Permanent Storage

When immediate program operations finish, datasets must be stored permanently. Unlike primary memory, secondary devices are completely non-volatile and offer immense storage limits.

- **Hard Disk Drives (HDD):** Traditional high-capacity magnetic platters used for bulk computing backup.
- **Optical Disks (CD / DVD):** Flat plastic disks read by a system laser beam, highly useful for backup.
- **Solid State Flash Media (USB / Pen Drives):** High-speed solid-state chips offering reliable portable data storage.

computer storage devices hard disk usb pen drive cd dvd

Above: Modern portable secondary media drives.

MASTER SHORTCUTS: A TO M

CLASS 10 • CS

System Efficiency (Part 1)

These operating shortcuts boost application speed and optimize layout tasks inside word processors, file explorers, and presentations.

Select All Assets	Ctrl + A	Bold Text Style	Ctrl + B	Copy Selected Item	Ctrl + C
Duplicate / Font Dialog	Ctrl + D	Align Center	Ctrl + E	Find Target Text	Ctrl + F
Go to Target Location	Ctrl + G	Find & Replace	Ctrl + H	Italicize Text Style	Ctrl + I
Justify Document Layout	Ctrl + J	Insert Hyperlink	Ctrl + K	Align Left Layout	Ctrl + L

💡 Important PowerPoint Note: Inserting a brand new presentation slide — Ctrl + M

MASTER SHORTCUTS: N TO Z

CLASS 10 • CS

System Efficiency (Part 2)

Complete operational table mapping keystrokes to optimize routine workflows and desktop window navigation.

Open New Document	Ctrl + N	Open Saved File	Ctrl + O	Print Current Page	Ctrl + P
Quit / Clear Paragraph	Ctrl + Q	Align Right Layout	Ctrl + R	Save Document Data	Ctrl + S
Open New Browser Tab	Ctrl + T	Underline Selected	Ctrl + U	Paste Clip Assets	Ctrl + V
Close Window / Tab	Ctrl + W	Cut Highlighted Data	Ctrl + X	Redo Last Canceled Step	Ctrl + Y
Undo Previous Action	Ctrl + Z	Interactive Zoom In	Ctrl + "+"	Interactive Zoom Out	Ctrl + "-"

INTERACTIVE LESSON QUIZ

CLASS 10 • CS

Verify Your Core Concepts

Test your operational hardware and execution knowledge against standard Class 10 assessment criteria.

Q1: Which logical hardware component of the Central Processor resolves computational inequalities (>, <, =)?

- A Control Unit (CU)
- B Arithmetic Logic Unit (ALU)
- C Main Memory Registers (MU)

Q2: Identify the character key combo programmed to launch a fresh operational slide inside MS PowerPoint.

- A Ctrl + N
- B Ctrl + M
- C Ctrl + S

ANY QUESTIONS?

Class 10 Complete Revision Series. Reference additional notes for hardware architecture diagrams.

WWW.TECH-SCHOOL.EDU • COMPUTER-DEPT@SCHOOL.COM

IMAGE SOURCES

Thumbnail for www.vecteezy.com

https://static.vecteezy.com/system/resources/previews/070/966/041/non_2x/modern-cpu-microprocessor-chip-technology-hardware-isolated-png.png

Source: www.vecteezy.com

Thumbnail for www.vecteezy.com

https://static.vecteezy.com/system/resources/previews/070/852/805/large_2x/close-up-of-a-computer-motherboard-with-electronic-components-and-circuitry-photo.jpg

Source: www.vecteezy.com

https://informationtechnologydit.wordpress.com/wp-content/uploads/2016/11/a.png?w=640

Source: informationtechnologydit.wordpress.com

Thumbnail for computerguidehub.com

https://computerguidehub.com/wp-content/uploads/output-devices-of-computer.webp

Source: computerguidehub.com

https://miro.medium.com/1*dmmp6M9NCdPJ14bA82pGLw.jpeg

Source: medium.com